

Define Problem Statements

Date	20 July 2026
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template)

Farmers often struggle to determine the most suitable crop based on soil nutrients and climate conditions, which can lead to financial losses and low agricultural yield. The Customer Problem Statement below captures this challenge from the farmer's point of view.

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A small/marginal farmer deciding what to grow this season	Choose the most profitable and suitable crop for my land	I don't have reliable, data-driven guidance on which crop fits my soil and weather conditions	I lack access to agronomists or soil-testing labs and rely on guesswork or tradition	Anxious about crop failure and financial loss
PS-2	A farmer managing limited water and fertilizer resources	Optimize the use of water, fertilizer, and soil nutrients	I over- or under-apply inputs without knowing the ideal nutrient levels for my crop	There is no easy tool that translates NPK/pH/climate readings into an actionable recommendation	Frustrated by wasted resources and inconsistent yields
PS-3	An agricultural extension officer supporting many farmers	Advise multiple farmers quickly and accurately	Manual, case-by-case assessment doesn't scale across large farmer populations	Data-driven decision support tools are not commonly deployed at the grassroots level	Overwhelmed trying to serve everyone effectively

Root Problem Summary

Modern agriculture faces poor crop selection, inefficient resource utilization, changing environmental conditions, and reduced farming productivity — largely because farmers and extension workers lack an accessible, intelligent, data-driven system to translate soil and climate readings into a concrete crop recommendation.